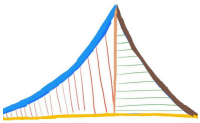
POMPS



G. V. V. Sharma

About this Book

This book introduces matrices through high school coordinate geometry. This approach makes it easier for beginners to learn Python for scientific computing. All problems in the book are from NCERT mathematics textbooks from Class 9-12. Exercises are from CBSE board exam papers.

The content is sufficient for industry jobs and covers nearly all matrix prerequisites for machine learning. There is no copyright, so readers are free to print and share.

License: https://creativecommons.org/licenses/by-sa/3.0/
and
https://www.gnu.org/licenses/fdl-1.3.en.html
First manual appeared in January 2018
First edition published on July 10, 2024
In this edition, some incorrect solutions were removed. All figures redrawn. More problems added.

Contents

1 Signal Processing 4

1 Signal Processing

1.0.1 A spring-mass-damper system with a mass of 1 kg is found to have a damping ratio of 0.2 and a natural frequency of 5 rad/s. The damping of the system is given by (AE 2007)

a) 2 Ns/m b) 2 N/s c) 0.2 kg/s d) 0.2 N/m

1.0.2 The Euler iteration formula for numerically integrating a first order nonlinear differential equation of form $x = f(x)$, with a constant step size of Δt is (AE 2007)

a) $x_{k+1} = x_k - \Delta t \times f(x_k)$
b) $x_{k+1} = x_k + \left(\Delta t^2/2\right) \times f(x_k)$
c) $x_{k+1} = x_k - (1/\Delta t) \times f(x_k)$
d) $x_{k+1} = x_k + \Delta t \times f(x_k)$